Tomato Soil pH Guide

What pH is, why it matters, how to measure it and how to change it

Quick answer: Tomatoes grow best at a soil pH of **6.0 to 6.8**. A pH of 6.8 is acceptable, but it is at the upper edge of the range. If your leaves are yellow, pH 6.8 is rarely the main cause. Check the pattern of yellowing, water, nutrition, pests and disease first.

1. What is pH?

pH measures how acidic or alkaline a soil is, on a scale from 0 to 14. Below 7 is acidic, 7 is neutral and above 7 is alkaline. pH is not a nutrient. It controls **how easily the plant can take up the nutrients** that are already in the soil.

2. Ideal pH for tomatoes

Growing method	Best pH range	Note
Soil (field, garden, raised bed)	6.0 to 6.8	6.2 to 6.5 is often called the sweet spot
Soilless media (coco, peat mixes), hydroponics	5.5 to 6.5	Lower range because there is no soil buffering

3. How pH affects the plant

pH level	What happens	Typical signs
Too low (below about 5.5-6.0)	Calcium and magnesium become less available. Aluminum and manganese can become toxic. Phosphorus can be locked up.	Slow growth, magnesium deficiency, blossom end rot, weak roots
Ideal (6.0 to 6.8)	Most nutrients are available at the same time.	Healthy growth and good yield
Too high (above about 7.0-7.5)	Iron, manganese, zinc, copper and boron are hard to take up. Phosphorus can be locked up.	Yellow young leaves with green veins (iron chlorosis), stunted plants

Tomatoes are less sensitive than some plants, such as blueberries, but a long-term pH problem still lowers yield.

4. Yellow leaves and pH 6.8

What you see	Is pH 6.8 likely the cause?	Look at first
Old, bottom leaves pale yellow	Unlikely	Nitrogen, over- or under-watering, natural ageing
Old leaves yellow between green veins	Unlikely	Magnesium deficiency
Young, top leaves yellow with green veins	Possible, especially in limy soil or hard water	Iron, manganese, zinc. Check drainage and test the soil
Yellow leaves with spots, curling, webs or mosaic	No	Diseases or pests (see the Tomato Disease Guide)

Hard (alkaline) irrigation water can slowly raise soil pH over time, even if your first reading looked fine.

5. How to measure pH correctly

- Take several samples from different spots, at root depth (about 10-20 cm), not only from the surface.
- Mix the samples, let the soil dry and test it. A laboratory test is the most reliable.
- Cheap strips and probes can be wrong by 0.5 pH units or more. Calibrate digital meters with buffer solutions.
- Test again after a few weeks. Soil pH changes slowly and is affected by fertilizer, water and compost.
- Also ask for EC (salt level) and nutrient levels. They are often more useful than pH alone.

6. How to change pH (only if you really need to)

Goal	Options	Notes
Lower pH (soil too alkaline)	Elemental sulfur; acid-forming fertilizers such as ammonium sulfate; compost or peat; acidify alkaline irrigation water carefully (for example in drip or hydroponic systems)	Sulfur works slowly (weeks to months). Do not over-acidify irrigation water.
Raise pH (soil too acidic)	Agricultural lime (calcium carbonate); dolomite lime if magnesium is also low; small amounts of wood ash	Lime also adds calcium, which helps against blossom end rot.

- Change pH slowly. A sudden big change can stress the roots.
- The amount you need depends on the soil. Clay and organic soils resist change, while sandy soils change quickly.
- Retest after 4-8 weeks before you add more.
- Never adjust pH based on one single reading. Confirm it first.

7. Simple checklist

- pH between 6.0 and 6.8? Then leave it alone.
- Look at which leaves are yellow (old or young) and how (whole leaf or between veins).
- Check water: soil moisture, drainage and roots.
- Look for spots, curling, webs, whiteflies and mites.
- Test the soil in a lab if the problem continues (pH, EC, nutrients).
- Change one thing at a time so you can see what worked.

Note: Amounts of lime, sulfur and fertilizer depend on your soil type and local conditions. Follow your soil test report and product labels, and ask a local agronomist when possible.